Acoustic and prosodic speech features as markers of depression, anxiety, stress, agitation, and apathy in adults: a systematic review of associations, elicitation paradigms, within speaker repeatability, and collection adherence

Megan Stubbs, Lexi Hartman, Shaba Rahavi, Zvika Shinar, Julian Thrash, Ann Hartman, Lovette Ochieng, Yitzi Devor, Sultan Scafi

PREVIEW

REVIEW TITLE AND BASIC DETAILS

Review title

Acoustic and prosodic speech features as markers of depression, anxiety, stress, agitation, and apathy in adults: a systematic review of associations, elicitation paradigms, within speaker repeatability, and collection adherence

Review type

No preview available

Condition or domain being studied

Depression; Depressive symptoms; Depressed Mood; Anxiety; Anxiety Disorder; Anxiety Symptoms; Stress; Agitated; Irritability; Irritation Symptom; Indifference/Apathy

Mental disposition in adults, comprising depression, anxiety, psychological stress, agitation or irritability, and apathy. These states are common, frequently under detected, and associated with poor outcomes, particularly in older adults and in long term care settings. The domain of interest is the measurement of these states through acoustic and prosodic properties of the voice, as a candidate route to passive, low burden monitoring.

Rationale for the review

Depression, anxiety, psychological stress, agitation, and apathy are common in adults and frequently under detected, particularly in older adults and long term care, where they predict functional decline and poorer outcomes. Detection relies on episodic self report and clinician observation, neither of which captures change within an individual between assessments.

Acoustic and prosodic speech features have been proposed as a passive, low burden signal. A 2025 systematic review by Martinez-Nicolas and colleagues synthesized 16 studies in older adults, finding speech rate, pause structure, and intensity consistent, and fundamental frequency inconsistent.

Four gaps remain. First, the evidence base has not been characterised by speech elicitation paradigm. Most work analyses spontaneous or interview speech recorded in laboratories, whereas passive monitoring yields short, fixed content utterances in uncontrolled environments. The proportion resting on the latter is unknown. Second, within speaker repeatability has not been synthesized alongside discriminative performance. Baseline referenced comparison requires knowing how far a feature varies absent any change in state; a feature with strong discrimination and poor repeatability would appear well supported yet be unusable. Third, states other than depression are rarely reviewed alongside it despite frequent co occurrence. Fourth, the evidence has not been characterised by whether people will provide recordings repeatedly over months. A feature that discriminates well and is stable is unusable if the protocol needed to obtain it is not one people sustain.

Registration is prospective so that the feature space and collection protocol are defined before primary data collection and subsequent analysis is confirmatory.

Review objectives

Primary objective. To identify and characterize acoustic and prosodic speech features reported to be associated with, or to detect, depression, anxiety, psychological stress, agitation or irritability, and apathy in adults, and to summarize the direction and consistency of those findings by feature. Secondary objectives. To characterize the speech elicitation paradigms on which that evidence rests, and to quantify the proportion of the evidence base using short fixed content speech. To characterize the contributing study designs, and specifically the proportion using longitudinal or repeated measures designs rather than cross sectional designs, and the proportion combining repeated within person measurement with short fixed content speech. To summarize reported within speaker repeatability of the identified features across separate recording occasions, expressed as intraclass correlation coefficients, coefficients of variation, standard error of measurement, minimal detectable change, or equivalent statistics. To characterize adherence to repeated voice collection, and to identify the collection modalities, prompting schedules, elicitation paradigms, and study durations associated with higher sustained adherence. This objective is not restricted to studies whose purpose is biomarker validation. To assess risk of bias using design matched appraisal tools, and to identify features whose supporting evidence is limited by study quality.

Keywords

Speech; Voice; Acoustic Analysis; Prosody; Digital biomarkers; Depression; Anxiety; Agitation; Apathy; Older adults; Reliability; Systematic review; Adherence; Compliance; Feasibility; Remote monitoring; Longitudinal data

Country

United States of America

ELIGIBILITY CRITERIA

Population

Included

Adults aged 18 and over, in any setting, including long term care. No restriction on country, comorbidity, or health status.

For the primary objective, studies must include participants in whom at least one target state (depression, anxiety, psychological stress, agitation or irritability, apathy) has been assessed using a validated instrument, clinician rating, or diagnosis. Clinical and non clinical samples are eligible, including unselected samples reporting symptom severity.

For the secondary repeatability objective, studies must include adults healthy or clinically stable across the measurement window.

For the secondary adherence objective, studies must include adults from whom voice or speech recordings were requested on two or more separate occasions, and must report a measure of adherence, completion, retention, attrition, or usable data yield. No target mental state is required. Eligibility here is deliberately wider: samples selected on the basis of dementia, mild cognitive impairment, delirium, Parkinson disease, stroke, or laryngeal pathology are eligible, since acoustic confounding does not bear on adherence, and these populations are directly relevant to applicability.

Mixed age samples are eligible where adult data are reported separately or obtainable from authors. Mixed clinical samples are eligible where results for a target state are separable.

Excluded

Children and adolescents under 18 years of age, and mixed age samples in which adult data cannot be separated. This exclusion applies to all objectives.

The following clinical exclusions apply to the primary and repeatability objectives only, and do not apply to the adherence objective. Samples selected on the basis of dementia, mild cognitive impairment, delirium, psychosis or schizophrenia.

Samples selected on the basis of a neurological or structural condition expected to alter acoustic measures independently of mental state, including Parkinson disease, stroke with dysarthria or aphasia, and laryngeal pathology. Participants with these conditions occurring incidentally within a broader sample do not trigger exclusion, provided the sample was not selected on that basis.

For the repeatability objective, samples undergoing treatment, acute illness, or other change in state between recording occasions.

For the adherence objective, studies reporting only overall study retention or trial attrition, with no denominator specific to requested voice or speech recordings.

Intervention(s) or exposure(s)

Included
For the primary and repeatability objectives, the exposure is quantitative acoustic or prosodic measurement of recorded voice or speech. Studies must extract at least one feature by automated or semi automated analysis. Eligible features include, but are not limited to, fundamental frequency and its variability, jitter, shimmer, harmonics to noise ratio, formant and derived articulatory measures, intensity, cepstral measures including CPP and MFCCs, speech and articulation rate, pause frequency and duration, phonation time, and standardized sets such as eGeMAPS.

Speech may be elicited by any paradigm. Paradigm is not an eligibility criterion; it is extracted from every study and used as the primary stratification variable. Recordings may be made in any setting, using any equipment, at any sampling rate. Extraction software and parameter settings are recorded, since acoustic values are not comparable across implementations.

For the adherence objective, the exposure is a protocol for repeated collection of recordings from the same participant across two or more occasions, by any modality, including smartphone, telephone or interactive voice response, smart speaker or embedded device, wearable, and researcher administered recording. Acoustic feature extraction is not required. Collection modality, prompting method, requested frequency and duration, per occasion task, and supervision are extracted.

Excluded
Studies using acted, simulated, or portrayed emotion, including performances by actors and standard emotional speech corpora recorded under instruction. These do not reflect naturally occurring mental states, and recordings produced under instruction do not inform voluntary participant adherence. This exclusion applies to all objectives.

The following exclusions apply to the primary and repeatability objectives only, and do not apply to the adherence objective. Studies analyzing only linguistic, lexical, semantic, or syntactic content, with no acoustic or prosodic measurement. Studies reporting only perceptual or listener based ratings of voice, with no quantitative acoustic extraction. Studies of non speech vocalizations only, such as cough, breath, or laughter, with no speech or phonation measure. Studies in which acoustic features are reported but not related to a target state or, for the repeatability objective, to repeatability across occasions.

For the adherence objective, studies in which recordings were captured without any requested participant action, since adherence is not a meaningful quantity where no recording was requested. Vocalization type and requested task are extracted from every study contributing to this objective.

Comparator(s) or control(s)

Included
Comparators vary by design. All of the following are eligible.

For case control and group comparison studies: participants without the target state, participants below a diagnostic threshold on a validated instrument, or healthy controls, assessed on the same acoustic measures under the same

elicitation conditions.

For diagnostic accuracy and classification studies: a validated instrument, structured clinical interview, clinician rating, or clinical diagnosis serving as the reference standard against which classification is evaluated.

For correlational and non-comparative studies: no comparator is required. The quantity of interest is the association between a feature and a continuous or ordinal measure of target state severity within a single sample.

For longitudinal and repeated measures studies: the same participants at a different time point, or during a contrasting state, for example before and after treatment, or symptomatic and remitted periods.

For the repeatability objective: the same speaker measured on a separate recording occasion, with no change in state between occasions.

For the adherence objective, no comparator is required. Single arm feasibility studies reporting an adherence proportion with a stated denominator are eligible. Where a study compares collection modalities, prompting schedules, or elicitation tasks, that comparison is extracted.

Excluded
The following exclusions apply to the primary and repeatability objectives only.

Comparisons in which the contrast group differs from the index group on a factor expected to alter acoustic measures independently of mental state, and where that factor is not accounted for. This includes comparisons across different elicitation tasks, different recording equipment or sampling rates, or substantially different age distributions, where the comparison is confounded by that difference rather than by the target state.

Comparisons against normative reference values drawn from a separate published sample, where no contemporaneous comparison group was recorded, unless the study also reports a within sample association with the target state.

Comparisons against acted, simulated, or portrayed emotional speech are excluded for all objectives.

For the adherence objective, comparisons across collection modalities, devices, or elicitation tasks are not excluded and are a pre specified point of interest, since the quantity compared is adherence rather than an acoustic value.

Study design
Both randomized and nonrandomized study types will be included.

Included
Primary objective: cross sectional, case control, cohort, longitudinal, and repeated measures observational designs; diagnostic test accuracy studies; and prediction model development or validation studies using acoustic features.

Repeatability objective: any design reporting a within speaker repeatability or measurement error statistic across two or more separate recording occasions, including test retest reliability studies and the control or stable arms of longitudinal studies.

Adherence objective: feasibility and pilot studies, usability studies, single arm observational studies, cohort and longitudinal studies, and the relevant arms of intervention studies, in each case where repeated voice collection was attempted and an adherence, completion, retention, attrition, or data yield measure is reported.

Randomized and non-randomized intervention studies are eligible where acoustic measures and a target state are reported for the same participants. From these, only the observational association between acoustic features and the target state is extracted. Treatment effects are not an outcome of this review, and randomization status has no bearing on eligibility.

Conference proceedings reporting original data are eligible. A substantial proportion of the acoustic and speech engineering literature appears in proceedings rather than journals, and excluding them would omit a meaningful part of the evidence base.

No restriction is placed on sample size.

Excluded
Case reports and case series.

Narrative reviews, systematic reviews, meta analyses, and scoping reviews. Reference lists of relevant reviews will be hand searched for eligible primary studies.

Editorials, commentaries, letters, and opinion pieces.

Conference abstracts without sufficient data for extraction, and study protocols without results.

Animal studies.

For the primary and repeatability objectives only: methodological or algorithm development papers that report classification performance on speech corpora without any clinical assessment of a target state in the participants. Such papers remain eligible for the adherence objective where they report repeated voice collection and an adherence measure.

Context
No restriction is placed on study setting. Studies conducted in laboratory, clinical, residential, or naturalistic and in situ settings are all eligible.

Recording context is not an eligibility criterion but is extracted from every included study and categorized as laboratory or sound treated environment, clinical or research setting under controlled conditions, or ecological capture in the participant's own living environment. Microphone type, recording device, sampling rate, and whether recording was researcher supervised are extracted where reported. Recording context is used in stratification and in the applicability assessment, since acoustic measures are sensitive to environment and equipment, and the transferability of laboratory derived findings to uncontrolled ambient capture is a stated concern of this review.

Publications in English are eligible. Studies published from 2005 onwards are eligible, on the grounds that automated acoustic feature extraction and the software implementations in common use were not widely established before that point, and that earlier work is not comparable in method to the current evidence base.

No restriction is placed on country, healthcare system, or funding source.

SIMILAR REVIEWS

Check for similar records already in PROSPERO
PROSPERO identified a number of existing PROSPERO records that were similar to this one (last check made on 9 August 2026). These are shown below along with the reasons given by that the review team for the reviews being

different and/or proceeding.

- Voice and Speech Analysis as an Early Diagnostic Tool for Parkinson's Disease [published 18 November 2025] [CRD420251184805]. The review was judged **not to be similar**
- Visual Text Support and EEG Markers of Spoken Language Comprehension in Healthy Adults: A Meta-Analysis [published 20 April 2026] [CRD420261374645]. The review was judged **not to be similar**
- Predictive performance of machine learning models using passive smartphone and wearable sensing to screen for depression, anxiety, and stress in adolescents and young adults: a systematic review. [published 18 November 2025] [CRD420251231371]. The review was acknowledged as **similar** but the authors opted to continue. The authors also noted: *"Similar in target states and in the broad framing of passive sensing for mental health detection, but distinct in population, exposure, and objectives. Population: that review covers adolescents and young adults aged 10 to 35. The present review covers adults aged 18 and over, with adults aged 60 and over as a pre specified subgroup, and excludes participants under 18. Exposure: that review covers passive sensing data from smartphones and wearables broadly, which typically comprises movement, location, screen and phone usage, sleep, and physiological signals. The present review is restricted to acoustic and prosodic features extracted from voice and speech. Objectives: that review evaluates predictive performance of machine learning models. The present review synthesizes associations as well as detection performance, and adds two objectives that review does not address: characterization of the evidence base by speech elicitation paradigm, and within speaker repeatability of acoustic features across separate recording occasions. Proceeding with registration."*

TIMELINE OF THE REVIEW

Date of first submission to PROSPERO
16 August 2026

Review timeline
Review start date: 9 August 2026. Review end date: 15 May 2027.

Date of registration in PROSPERO
17 August 2026

AVAILABILITY OF FULL PROTOCOL

Availability of full protocol
A full protocol has not been written.

SEARCHING AND SCREENING

Search for unpublished studies
Both published and unpublished studies will be sought.

Main sources that will be searched
The main sources to be searched are *CINAHL - Cumulative Index to Nursing and Allied Health Literature*, *Embase*, *PsycINFO*, *PubMed*, *SCIE - Science Citation Index Expanded* and *SSCI - Social Sciences Citation Index*.

Other sources that will be searched
IEEE Xplore: acoustic and speech engineering literature including Interspeech and ICASSP proceedings. ProQuest Dissertations and Theses: unpublished doctoral work. The medRxiv, PsyArXiv, and arXiv preprint servers, for preprints without a corresponding peer reviewed publication. Reference lists of all included studies and of relevant prior systematic reviews will be hand searched.

Search language restrictions
The review will only include studies published in English.

Search date restrictions
Databases will be searched for articles published from 1 January 2005, there are no search end date restrictions.

Other methods of identifying studies
Other studies will be identified by: *contacting authors or experts*, *looking through all the articles that cite the papers included in the review ("snowballing" or forward citation searching)*, *reference list checking (backward citation searching)*, *searching conference proceedings* and *searching dissertation and thesis databases*.

Additional information about identifying studies
Conference proceedings via IEEE Xplore, including Interspeech and ICASSP. Web of Science Core Collection with SCIE, SSCI, and CPCI-S enabled. Dissertations via ProQuest. Preprints via medRxiv, PsyArXiv, and arXiv, where no peer reviewed version exists. Forward citation searching via Web of Science and OpenAlex; Scopus is not institutionally available.

Link to search strategy
A full search strategy is not available.

Selection process
Studies will be screened independently by at least two people (or person/machine combination) with a process to resolve differences.

Other relevant information about searching and screening
Deduplication will be conducted in Rayyan supplemented by manual checking, with particular attention to two cases specific to this evidence base. First, conference proceedings frequently precede an extended journal version of the same work; where both are identified, the more complete version is used and the pair is recorded as a single study. Second, a small number of speech corpora are reused across many publications, so studies will be linked by underlying sample during extraction to prevent the same participants contributing more than once to the synthesis of any feature.

DATA COLLECTION PROCESS

Data extraction from published articles and reports
Data will be extracted independently by at least two people (or person/machine combination) with a process to resolve differences.

Authors will be asked to provide any required data not available in published reports.

Study risk of bias or quality assessment
Risk of bias will be assessed using: *JBI tools*, *PROBAST*, *QUADAS-2* and *QUAPAS*

Tools assigned by design, pre specified: JBI analytical cross sectional (primary); QUADAS-2, accuracy; PROBAST, prediction models; QUAPAS, prognostic accuracy; COSMIN reliability box, repeatability; JBI prevalence, adherence proportions; MMAT, mixed methods usability. COSMIN and MMAT are not in the tool list above. Reported stratified by design, not aggregated.

Data will be assessed independently by at least two people (or person/machine combination) with a process to resolve differences.

Additional information will be sought from study investigators if required information is unclear or unavailable in the study publications/reports.

Reporting bias assessment
Studies often extract large feature sets and report only significant results. Features stated as extracted are compared against those reported, and unreported features recorded. For the adherence objective, whether a completion figure and denominator are reported is recorded. Protocols checked where available. Narrative and study level; no funnel plots.

Certainty assessment
Certainty assessed per feature, not per study. Modified GRADE for observational associations, considering risk of bias, inconsistency, indirectness, imprecision, and reporting bias. Indirectness includes elicitation paradigm and recording

context. Repeatability evidence assessed using COSMIN modified GRADE. Features graded separately for association evidence and for within speaker stability.

Certainty for the adherence objective is assessed narratively by collection modality rather than by acoustic feature, considering risk of bias, inconsistency, indirectness, and imprecision, with indirectness covering population, setting, and collection duration.

OUTCOMES TO BE ANALYSED

Main outcomes

Primary outcome: the relationship between an acoustic or prosodic speech feature and a target state of mental disposition, comprising depression, anxiety, psychological stress, agitation or irritability, and apathy.

The target state must be established using a validated instrument, structured clinical interview, clinician rating, or clinical diagnosis. Acceptable instruments include, but are not restricted to, the Geriatric Depression Scale, PHQ-9, Beck Depression Inventory, Hamilton Rating Scale for Depression, Montgomery-Asberg Depression Rating Scale, GAD-7, State-Trait Anxiety Inventory, Hospital Anxiety and Depression Scale, Perceived Stress Scale, Cohen-Mansfield Agitation Inventory, Neuropsychiatric Inventory, and Apathy Evaluation Scale. Diagnoses established against DSM or ICD criteria are acceptable.

Acoustic features are quantitative measures extracted from recorded voice by automated or semi automated analysis, including fundamental frequency and variability, jitter, shimmer, harmonics to noise ratio, formant measures, intensity, cepstral measures, speech rate, articulation rate, and pause timing measures.

Effect measures: correlation coefficients, standardized mean differences, regression coefficients, and, for detection studies, sensitivity, specificity, area under the curve, and accuracy. Reported as presented; no pooling is planned.

Time points: any. Cross sectional measurement is eligible. Where acoustic measurement and target state assessment occur at different time points, the interval is extracted. Where repeated measurement occurs, all time points are extracted and the study is flagged as longitudinal.

Outcomes are synthesized by acoustic feature rather than by study, so that each feature carries its own body of evidence, direction, consistency, and certainty rating.

Additional outcomes

Additional outcome 1: composition of the evidence base by speech elicitation paradigm, classified as sustained vowel or syllable phonation; isolated words or phrase repetition; read passages or standardized texts; picture description or structured task; or spontaneous conversation and clinical interview. Utterance duration and whether content was fixed are also extracted. Effect measure: counts and proportions of studies and participants by paradigm, cross tabulated by target state and feature. The proportion using short fixed content speech is the quantity of primary interest.

Additional outcome 2: composition by study design and measurement timing, classified as cross sectional, case control, cohort, longitudinal, repeated measures, or intervention. Effect measure: counts and proportions by design, and the proportion providing within person repeated measurement. Longitudinal and repeated measures designs are a pre specified priority stratum. Within it, studies combining repeated within person measurement with short fixed content speech form a pre specified subgroup. A small or empty count is itself a substantive finding.

Additional outcome 3: within speaker repeatability, the stability of a feature measured on the same speaker across two or more separate recording occasions in adults healthy or clinically stable across the window. Acceptable measures: intraclass correlation coefficient, coefficient of variation, standard error of measurement, minimal detectable change, limits of agreement, repeatability coefficient. Interval between occasions is extracted. Reported by feature alongside the association evidence.

Additional outcome 4: adherence to repeated voice collection, the extent to which adults provide recordings as scheduled across two or more occasions. Acceptable measures: adherence, completion or response rate; recordings

provided against expected; retention; attrition; usable data yield; reported burden or acceptability. Effect measure: proportions with denominators, reported by collection modality, prompting schedule, elicitation paradigm, study duration, and population.

PLANNED DATA SYNTHESIS

Strategy for data synthesis

Structured narrative synthesis without meta-analysis, reported per SWiM. Pooling is not planned: heterogeneity in target states, instruments, elicitation paradigms, feature definitions, and extraction software makes pooled estimates uninterpretable, and the review questions do not require them. Synthesis is organized by acoustic feature rather than by study. For each feature, the review reports the number of contributing studies and participants, target states examined, elicitation paradigms and recording contexts represented, the direction and consistency of reported associations, the risk of bias profile of contributing studies, and a certainty rating. Standardization of effect direction is the grouping method; effect direction plots and harvest plots are used in preference to vote counting on statistical significance. Where effect estimates are comparable in form, ranges are reported rather than pooled. Descriptive counts and cross tabulations are used for the elicitation paradigm and study design outcomes across the full included set. Repeatability evidence is synthesized separately by feature and presented alongside the association evidence for the same feature, so that discriminative performance can be read against within speaker stability. If a sufficiently homogeneous subset emerges, defined as three or more studies sharing target state, elicitation paradigm, feature definition, and effect measure, meta-analysis of that subset may be considered as a post hoc addition and would be reported as such. Extraction is dual independent throughout and tiered by depth, not by reviewer count. Studies using short fixed content speech, comprising sustained vowels or syllables, isolated words or phrase repetition, and read passages or standardized texts, receive full extraction and form the primary synthesis. Studies using picture description, structured interview, or spontaneous conversation receive a reduced six field extraction and supply the denominator for the elicitation paradigm and study design outcomes. Studies extracted at reduced depth are not risk of bias appraised, which is pre specified here and reported as a limitation.

Adherence evidence is synthesized separately and descriptively. Adherence proportions are reported with their denominators and tabulated by collection modality, prompting schedule, elicitation paradigm, study duration, and population, with ranges rather than pooled estimates. Studies contributing only to the adherence objective receive a targeted extraction covering collection protocol, adherence measure, and population, and are appraised using the JBI prevalence checklist, or MMAT where the study is mixed methods. The intersection of high sustained adherence with short fixed content speech is a pre specified point of interest, since it corresponds most directly to passive monitoring by an embedded device.

LIVING REVIEW METHODS

Living systematic review methods

No preview available

CURRENT REVIEW STAGE

Stage of the review at this submission

Review stage	Started	Completed
Pilot work	✓	✓
Formal searching/study identification	✓	✓
Screening search results against inclusion criteria	✓	✓
Data extraction or receipt of IPD	✓	
Risk of bias/quality assessment	✓	

Review stage	Started	Completed
Data synthesis	✓	

Review status
The review is currently planned or ongoing.

Publication of review results
Results of the review will be published in English.

REVIEW AFFILIATION, FUNDING AND PEER REVIEW

Review team members
Megan Stubbs (review guarantor and contact) ORCID: 0009-0001-8742-2024. University of North Dakota. United States of America.

Conflict of interest
Intellectual interest declared. The review team is developing the embedded speech monitoring device this review informs, and the review defines its acoustic feature space and voice collection protocol. Mitigations: prospective registration before any searching or screening; pre specified eligibility criteria and outcomes; and dual independent screening and extraction. No financial interest.

Lexi Hartman. University of North Dakota. United States of America.

Conflict of interest
Intellectual interest declared. The review team is developing the embedded speech monitoring device this review informs, and the review defines its acoustic feature space and voice collection protocol. Mitigations: prospective registration before any searching or screening; pre specified eligibility criteria and outcomes; and dual independent screening and extraction. No financial interest.

Shaba Rahavi. University of North Dakota. United States of America.

Conflict of interest
Intellectual interest declared. The review team is developing the embedded speech monitoring device this review informs, and the review defines its acoustic feature space and voice collection protocol. Mitigations: prospective registration before any searching or screening; pre specified eligibility criteria and outcomes; and dual independent screening and extraction. No financial interest.

Dr Zvika Shinar. ORCID: 0009-0006-2463-2872. University of North Dakota. United States of America.

Conflict of interest
Financial and intellectual interest declared. Dr Shinar is a shareholder in EchoWell, a company developing speech based monitoring technology which may benefit from this review. The review team is developing the device this review informs, and the review defines its acoustic feature space and voice collection protocol. Mitigations: prospective registration before any searching or screening; pre specified eligibility criteria and outcomes; dual independent screening and extraction; and exclusion of Dr Shinar from screening, extraction, risk of bias assessment, and arbitration.

Julian Thrash. University of North Dakota. United States of America.

Conflict of interest
Intellectual interest declared. The review team is developing the embedded speech monitoring device this review informs, and the review defines its acoustic feature space and voice collection protocol. Mitigations: prospective registration before any searching or screening; pre specified eligibility criteria and outcomes; and dual independent screening and extraction. No financial interest.

Ann Hartman. University of North Dakota. United States of America.

Conflict of interest
Intellectual interest declared. The review team is developing the embedded speech monitoring device this review informs, and the review defines its acoustic feature space and voice collection protocol. Mitigations: prospective

registration before any searching or screening; pre specified eligibility criteria and outcomes; and dual independent screening and extraction. No financial interest.

Lovette Ochieng. University of North Dakota. United States of America.

Conflict of interest
Intellectual interest declared. The review team is developing the embedded speech monitoring device this review informs, and the review defines its acoustic feature space and voice collection protocol. Mitigations: prospective registration before any searching or screening; pre specified eligibility criteria and outcomes; and dual independent screening and extraction. No financial interest.

Yitzi Devor. University of North Dakota. United States of America.

Conflict of interest
Intellectual interest declared. The review team is developing the embedded speech monitoring device this review informs, and the review defines its acoustic feature space and voice collection protocol. Mitigations: prospective registration before any searching or screening; pre specified eligibility criteria and outcomes; and dual independent screening and extraction. No financial interest.

Sultan Scafi. University of North Dakota. United States of America.

Conflict of interest
Intellectual interest declared. The review team is developing the embedded speech monitoring device this review informs, and the review defines its acoustic feature space and voice collection protocol. Mitigations: prospective registration before any searching or screening; pre specified eligibility criteria and outcomes; and dual independent screening and extraction. No financial interest.

Named contact
Megan Stubbs (megan.rossi@und.edu). ORCID: 0009-0001-8742-2024. University of North Dakota. United States of America.

Review affiliation
University of North Dakota

Funding source
Review has no specific/external funding but is supported by guarantor/review team (non-commercial) institutions.

Peer review
There has been no peer review of this planned review.

ADDITIONAL INFORMATION

Additional information
A systematic review of speech analysis for detecting depression in older adults was published in December 2025 by Martinez-Nicolas and colleagues in Frontiers in Psychology, including 16 studies in adults over 55. The present review is distinguished by a broader set of target states, an unrestricted adult population with adults aged 60 and over as a pre specified subgroup, characterization of the evidence base by elicitation paradigm, and secondary objectives on within speaker repeatability and on adherence to repeated voice collection. That review also seeds validation of search sensitivity.

This review will inform the acoustic feature specification and the voice collection protocol for an embedded speech monitoring device under development within the parent project, and the pre specified analysis plan for that device's evaluation. All review team members belong to that parent project. One member holds equity in EchoWell, a company developing speech based monitoring technology. These dependencies are declared under conflicts of interest.

Industry support for the parent project is pending an ARPA-H funding decision at the date of registration. This review holds no external funding as of that date. If funding is confirmed and extends to this review, the funding field will be amended and the change recorded in the version history.

Preliminary scoping searches have estimated record yield and validated terms against seed studies. No records have been collected and no study selection has taken place.

The present review also adds a third objective, on adherence to repeated voice collection, not addressed by any similar registered review.

Review conflict of interest

Declared individual interests are recorded under team member details. Nine review team members have declared a potential conflict of interest.This review also notes the following interests:

All four review team members are members of the parent project developing the embedded speech monitoring device that this review informs. One team member, Dr Zvika Shinar, additionally declares a financial interest as a shareholder in EchoWell, a company developing speech based monitoring technology which may benefit from the findings of this review. To mitigate potential bias, the review is prospectively registered before any formal searching or screening, eligibility criteria and outcomes are pre specified, screening and data extraction are conducted independently by at least two reviewers, and the team member holding the financial interest takes no part in screening, data extraction, risk of bias assessment, or arbitration of reviewer disagreements. The guarantor holds no financial interest. The review holds no external funding at the date of registration.

Medical Subject Headings

Depression; Depressive Disorder; Depressive Disorder, Major; Anxiety; Stress, Psychological; Psychomotor Agitation; Irritable Mood; Apathy; Speech Acoustics; Speech Production Measurement; Voice Quality; Phonation; Sound Spectrography; Voice; Aged; Adult; Long-Term Care; Homes for the Aged; Reproducibility of Results; Biomarkers; Patient Compliance; Treatment Adherence and Compliance; Feasibility Studies; Patient Dropouts; Ecological Momentary Assessment; Telemedicine

Revision note

Four new team members were added to the review.

PROSPERO version history

- Version 1.0, published 17 Aug 2026

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